

PASSES-HGV: Expanding the Continuous Spectral Framework to 6-DOF Hypersonic Glide and Fractional Orbital Trajectories

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Abstract

This paper presents PASSES-HGV, an open-source, continuous mathematical framework that expands the original fixed-grid spectral formulation into a full 6-Degree-of-Freedom (6-DOF) hypersonic glide and Fractional Orbital Bombardment System (FOBS) architecture. By replacing traditional discrete finite element methods and dense Chebyshev derivative operators with an Ultraspherical Spectral Method over a bivariate tensor-product space, the framework maintains C^∞ continuity and $O(1)$ operator conditioning across complex waverider geometries. Anisotropic flexural rigidity and localized torsional flutter are governed continuously using Mindlin-Reissner plate mechanics, while a coupled aerothermodynamic model links oblique shock-expansion theory, real-gas convective heating, and a Stefan moving-boundary ablation PDE directly to the structural state vector. To ensure autonomous trajectory optimization, an embedded Successive Convexification (SCvx) algorithm utilizes exact virtual control penalties to navigate thermal and skip-out constraints. Finally, the integration architecture handles a continuous physics "idle" during the Karman line transition, propagating a J_2 -perturbed Keplerian orbital coast before re-engaging quaternion kinematics for the terminal hypersonic reentry phase—all within a single, unbroken, GPU-accelerated numerical integration run.

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1 Introduction

The modeling and simulation of Hypersonic Glide Vehicles (HGVs) and Fractional Orbital Bombardment Systems (FOBS) present a highly coupled, multi-physics challenge that fundamentally strains traditional computational aerospace methods. Legacy trajectory and aeroelastic analysis often relies on discrete Finite Element Methods (FEM) and rigidly decoupled aerodynamic look-up tables. While effective for localized, static structural validation, these discrete methodologies introduce discontinuous boundary conditions, require mid-flight matrix inversions, and mandate prohibitively long computational wall-times.

This manuscript introduces PASSES-HGV, a comprehensive expansion of the original spectral multi-physics framework. By elevating the core fixed-grid architecture to a full 6-Degree-of-Freedom (6-DOF) kinematic environment, this framework establishes a continuous mathematical pipeline capable of seamlessly simulating lifting-body aerodynamics and exo-atmospheric orbital mechanics. To circumvent the severe $O(N^4)$ matrix ill-conditioning inherent to dense Chebyshev collocation [7], the HGV structural and thermodynamic domains are mapped using an Ultraspherical Spectral Method. The complex, direction-dependent torsional and transverse bending modes are governed continuously by the anisotropic structural rigidity(?) tensor, ensuring an exact, C^∞ differentiable representation of anisotropic flexural rigidity across the planform.

Furthermore, this framework integrates a fully autonomous hypersonic aerothermodynamic model. By coupling oblique shock-expansion approximations, Sutton-Graves heating, and Stefan moving-boundary pyrolytic recession directly to the structural state vector, the right-hand side operators remain perfectly continuous and suitable for massive GPU acceleration. Navigational control is achieved via an embedded Successive Convexification algorithm [5], allowing the vehicle to actively manage kinetic energy while autonomously respecting thermal limits. Ultimately, PASSES-HGV demonstrates the capability to execute an unbroken numerical integration spanning from atmospheric boost, through a J_2 -perturbed fractional orbit, and into the extreme aerothermal regime of hypersonic reentry.

2 The Expanded Global State Vector

2.1 The 6-DOF Attitude State

To prevent singularity issues (gimbal lock) when the HGV executes aggressive, near-vertical dive maneuvers or exo-atmospheric tumbles, we parameterize the vehicle's attitude using a unit quaternion $\mathbf{q} = [q_0, q_1, q_2, q_3]^T$. The time evolution of the attitude is coupled to the body-frame angular velocity $\boldsymbol{\omega}_B$ via the quaternion kinematic differential equation, which is integrated concurrently with the structural state $\mathbf{w}$ within the RKF45 solver.

2.2 Local Incidence Angle Mapping

Because the waverider has complex curvature and continuous aeroelastic deformation, the global angle of attack α is insufficient. The local incidence angle is thus the dot product of the normalized body-frame velocity vector and the deformed surface normal:

$$\sin(\delta_c(\xi, \eta)) = \mathbf{n}(\xi, \eta, t) \cdot \frac{\mathbf{v}_B}{|\mathbf{v}_B|}$$

2.3 Constructing the RHS Forcing Vector

Relying exclusively on modified Newtonian pressure is physically insufficient for slender waveriders at moderate angles of attack, as it ignores leeward suction and viscous interaction. Therefore, PASSES-HGV augments the Newtonian formulation with local oblique shock-expansion theory.

For the windward nodes, local pressure coefficients are continuously scaled by the freestream dynamic pressure to populate the quasi-static RHS forcing vector:

$$\mathbf{Q}_{aero} = q_{dyn} \mathbf{C}_p$$

This couples the rigid-body 6-DOF kinematics, the freestream environment, and the flexible structural dynamics into one continuous mathematical pipeline.

3 High-Fidelity Hypersonic Aerothermodynamics

3.1 Real-Gas Convective Heating: The Fay-Riddell Formulation

We anchor the continuous thermal model using the Fay-Riddell formulation for the stagnation point convective heat flux [2]. This accounts for boundary layer equilibrium chemistry and surface catalytic recombination:

$$\dot{q}_{s,conv} = 0.763 \text{Pr}^{-0.6} (\rho_e \mu_e)^{0.4} (\rho_w \mu_w)^{0.1} \sqrt{\left(\frac{du_e}{dx} \right)_s} (h_{0e} - h_w) \left[1 + (Le^\beta - 1) \frac{h_D}{h_{0e}} \right]$$

3.2 The Continuous 2D Total Heat Flux Tensor

The total stagnation heating $\dot{q}_s$ is mapped across the entire bivariate grid using the incidence angle $\delta_c(\xi, \eta)$. For the windward nodes ($\delta_c > 0$), a continuous modified Lees' distribution [4] governs the local heat flux:

$$\dot{q}_{total}(\xi, \eta) = \dot{q}_s \sin(\delta_c(\xi, \eta)) \left(\frac{R_{eff}}{x(\xi)} \right)^{0.5}$$

3.3 Moving-Boundary Stefan Energy Balance and Pyrolytic Ablation

While surface heat fluxes define the incoming thermal load, evaluating structural survivability requires explicitly tracking the geometric surface recession ($\dot{s}$). We introduce a moving-boundary Stefan partial differential equation to govern this thermochemical mass loss. The energy balance equation directly couples the incident heating, radiative cooling, and internal conduction to the material's ablation enthalpy (ΔH_{abl}):

$$\rho_m \Delta H_{abl} \dot{s}(x, y, t) = \dot{q}_{total}(x, y, t) - \epsilon \sigma T_{wall}^4 - k_{mat} \left. \frac{\partial T}{\partial n} \right|_{wall}$$

This continuous recession rate dynamically alters the local effective nose radius R_{eff} and updates the deformed surface normal vectors $\mathbf{n}(\xi, \eta, t)$.

4 Structural Generality: Mindlin-Reissner Anisotropic Plate Theory

Transitioning to a hypersonic lifting body means the vehicle is no longer governed by simple 1D beam mechanics. Furthermore, classical Kirchhoff-Love thin-plate theory ignores transverse shear deformations, which become the dominant mode of aeroelastic flutter in the thicker sections of the hull. We elevate the structural framework to a generalized Mindlin-Reissner anisotropic plate formulation [6].

4.1 The Expanded 3-DOF Structural State

The continuous structural state at any spatial coordinate expands into a 3×1 vector:

$$\mathbf{u}(\xi, \eta, t) = \begin{bmatrix} w \\ \phi_x \\ \phi_y \end{bmatrix}$$

Because the rotations are no longer locked to the gradient of the surface, the localized transverse shear strains are defined continuously across the domain as:

$$\gamma_{xz} = \phi_x + \frac{\partial w}{\partial x}, \quad \gamma_{yz} = \phi_y + \frac{\partial w}{\partial y}$$

4.2 Ultraspherical Block-Kronecker Stiffness Matrix

Dense Chebyshev differentiation matrices exhibit condition numbers scaling as $O(N^4)$, causing catastrophic precision loss for fourth-order bending operators. PASSES-HGV remediates this by implementing the Ultraspherical Spectral Method [7], converting the differential operators into sparse, banded matrices and preserving $O(1)$ operator conditioning.

By formulating the Mindlin-Reissner governing equations as a system of coupled, block-sparse Kronecker operators [1], the structural domain bypasses the discontinuous assembly steps of traditional FEM, enabling highly parallelized GPU evaluations.

5 Boundary Conditions

5.1 Anisotropic Rigidity Tensor

The hypersonic glide vehicle relies on a blended aerodynamic shell whose resistance to bending and torsion is governed by the spatially continuous anisotropic flexural rigidity tensor, $\mathbf{D}(\xi, \eta)$.

5.2 Mindlin-Reissner Free-Free Boundary Conditions

Applying classical Kirchhoff effective shear boundary conditions to a Mindlin-Reissner state space over-constrains the plate perimeter and induces artificial boundary-layer shear locking. Therefore, the free-free constraints are strictly formulated using three independent boundary equations for the three independent fields.

Along a free longitudinal edge, the zero bending moment constraint is mapped by the anisotropic structural rigidity(?) tensor:

$$M_x = \mathbf{D}_{11} \frac{\partial \phi_x}{\partial x} + \mathbf{D}_{12} \frac{\partial \phi_y}{\partial y} = 0$$

The zero twisting moment constraint requires:

$$M_{xy} = \mathbf{D}_{66} \left(\frac{\partial \phi_x}{\partial y} + \frac{\partial \phi_y}{\partial x} \right) = 0$$

And the zero transverse shear force is evaluated independently of the geometric slope:

$$Q_x = \kappa^2 G_{xz} h \left(\phi_x + \frac{\partial w}{\partial x} \right) = 0$$

6 GNC: Autonomous Successive Convexification

6.1 Linearization, Virtual Controls, and SOCP

To navigate the thermodynamic constraints, we reformulate trajectory generation as a constrained Optimal Control Problem using Successive Convexification (SCvx) [5]. The algorithm linearizes the non-linear dynamics around a known reference trajectory using a first-order Taylor expansion. However, linearizing non-convex dynamics frequently induces artificial infeasibility. To guarantee deterministic solver convergence, we introduce non-negative virtual control vectors $\mathbf{v}_i \in \mathbb{R}^{n_x}$ into the discretized constraints:

$$\mathbf{x}_{i+1} = \mathbf{A}_i \mathbf{x}_i + \mathbf{B}_i \mathbf{u}_i + \mathbf{z}_i + \mathbf{v}_i$$

To ensure these virtual controls are not physically commanded, an exact penalty term $w_{virt} \sum_{i=1}^{N-1} \|\mathbf{v}_i\|_1$ is added to the objective cost function. Furthermore, a dynamic trust-region bounds the step size based on the ratio of actual non-linear cost reduction to predicted convex cost reduction:

$$\rho_k = \frac{J(\mathbf{x}^{(k)}) - J(\mathbf{x}^{(k+1)})}{\Delta L^{(k)}}$$

6.2 Plasma Sheath Ionization and Dynamic EKF Gating

During atmospheric entry, the plasma sheath reflects electromagnetic signals, severing external navigation updates. To maintain state estimation during RF blackout, the framework utilizes an ionization-aware Extended Kalman Filter (EKF) [3].

When the evaluated plasma frequency exceeds the transmission frequency ($\omega_p \geq \omega_{GNSS}$), the EKF autonomously zeroes the corresponding rows in the observation matrix ($\mathbf{H}_k \rightarrow \mathbf{0}$). During this blackout, the state covariance $\mathbf{P}(t)$ balloons quadratically due to unobservable IMU biases. This expanding error footprint is actively evaluated by the SCvx algorithm, autonomously triggering pull-up maneuvers to extinguish the plasma sheath and force a GNSS re-acquisition window.

7 Orbital Mechanics and FOBS

7.1 J_2 Perturbed Orbital Mechanics

During a fractional orbit, a spherical gravity model is insufficient. We must update the kinematics to include the J_2 geopotential harmonic to account for Earth oblateness [8]. In the Earth-Centered Inertial (ECI) Cartesian frame, the gravitational acceleration vector $\mathbf{g}(\mathbf{r})$ becomes:

$$\mathbf{g}(\mathbf{r}) = -\frac{\mu}{|\mathbf{r}|^3} \mathbf{r} + \mathbf{g}_{J_2}(\mathbf{r})$$

This physically rigorous addition evaluates the secular trajectory variations, preventing catastrophic crossrange errors during reentry.

8 Conclusion

By anchoring the formulation upon Ultraspherical continuous operators and Mindlin-Reissner mechanics, PASSES-HGV delivers an unprecedented, highly robust mathematical architecture for next-generation multi-physics hypersonic simulation and defense engineering.

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